Law of Sines and Law of Cosines

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Law of Sines**

In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:

a angle C $= 75°$

b side b (2 d.p.) $= \frac{12\sin 65°}{\sin 40°} \approx 16.92$

c side c (2 d.p.) $= \frac{12\sin 75°}{\sin 40°} \approx 18.03$

(a) $= 75°$; (b) $= \frac{12\sin 65°}{\sin 40°} \approx 16.92$; (c) $= \frac{12\sin 75°}{\sin 40°} \approx 18.03$

2 **[Foundational]** • **Target: Law of Sines**

Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.

The angle at the balloon is $180° - 58° - 47° = 75°$. By the Law of Sines, $d = \frac{400\sin 47°}{\sin 75°} \approx 303$ m.

3 **[Foundational]** • **Target: Law of Cosines**

In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.

$c^2 = 8^2 + 11^2 - 2(8)(11)\cos 52° \approx 185 - 108.36 = 76.64$, so $c \approx 8.75$.

4 **[Foundational]** • **Target: Law of Cosines**

A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.

The largest angle is opposite the side of length 12: $\cos C = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126} = -\frac{1}{9}$, so $C \approx 96.4°$.

5 **[Foundational]** • **Target: Law of Cosines**

Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.

$A = \frac{1}{2}(10)(14)\sin 35° \approx 70 \times 0.5736 \approx 40.15$ square units.

6 **[Foundational]** • **Target: Law of Cosines**

A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

The angle between the legs is $90° + 50° = 140°$. Law of Cosines:

$d^2 = 30^2 + 45^2 - 2(30)(45)\cos 140° \approx 2925 + 2068.4 = 4993.4$, so $d \approx 71$ km.

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7 [Foundational] • Target: The ambiguous case (SSA)

In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

$\sin B = \frac{b\sin A}{a} = \frac{9(0.5)}{5} = 0.9$. This gives $B \approx 64.2°$ or $B \approx 115.8°$. Since $A + B < 180°$ in both cases, there are two possible triangles.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The ambiguous case (SSA)

In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

$\sin E = \frac{e\sin D}{d} = \frac{6\sin 40°}{10} \approx 0.386$, giving $E \approx 22.7°$ or $E \approx 157.3°$. Since $d > e$ (the side opposite the given angle is longer), only the acute solution is valid: exactly one triangle exists.

9 [Intermediate] • Target: Choosing the right law

For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

a Two angles and one side are known (AAS). Law of Sines — a full angle-side ratio pair is available.

b Three sides are known (SSS). Law of Cosines — no angle is known to pair with a side.

c Two sides and the included angle are known (SAS). Law of Cosines — the included angle lets you find the third side directly.

(a) Law of Sines — a full angle-side ratio pair is available.; (b) Law of Cosines — no angle is known to pair with a side.; (c) Law of Cosines — the included angle lets you find the third side directly.

10 [Intermediate] • Target: Choosing the right law

A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.

Find angle C between the sides of length 120 and 150 (opposite the side 90):
$\cos C = \frac{120^2 + 150^2 - 90^2}{2(120)(150)} = \frac{28800}{36000} = 0.8$, so $C \approx 36.87°$. Area
$= \frac{1}{2}(120)(150)\sin(36.87°) \approx 9000(0.6) = 5400$ m^2.

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11 [Intermediate] • Target: Heron's formula

State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.

$A = \sqrt{s(s-a)(s-b)(s-c)}$. Here $s = \frac{13+14+15}{2} = 21$. $A = \sqrt{21(8)(7)(6)} = \sqrt{7056} = 84$.

12 [Intermediate] • Target: Heron's formula

A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.

$s = \frac{9+12+15}{2} = 18$. $A = \sqrt{18(9)(6)(3)} = \sqrt{2916} = 54$ m^2. Check: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$ ✓ — confirms a right triangle, and indeed $\frac{1}{2}(9)(12) = 54$ matches.

13 [Intermediate] • Target: Navigation and bearing problems

A plane flies on a bearing of $N\,60°\,E$ for 200 km, then changes course to a bearing of $S\,80°\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.

The angle between the two bearings: turning from $N60°E$ to $S80°E$ sweeps through $180° - 60° - 80° = 40°$ measured one way, but the interior angle of the path (between the two flight segments) is $180° - 40° = 140°$. Displacement: $d^2 = 200^2 + 150^2 - 2(200)(150)\cos(140°)$.

14 [Intermediate] • Target: Solving oblique triangles from a diagram

In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.

$z^2 = 8^2 + 10^2 - 2(8)(10)\cos 48° \approx 164 - 107.0 = 57.0$, so $z \approx 7.55$. Then $\sin X = \frac{8\sin 48°}{7.55} \approx 0.787$, giving $X \approx 51.9°$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying triangle solutions with the angle sum • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

$Y = 180° - 48° - 51.9° = 80.1°$. Check: $48° + 51.9° + 80.1° = 180°$ ✓.

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16 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 7.04 Applications: Growth and Decay

A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.

(1) $\cos C = \frac{6^2 + 8^2 - 10^2}{2(6)(8)} = \frac{0}{96} = 0 \Rightarrow C = 90°$. (2) Since $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, this is a right triangle by the Pythagorean theorem, confirming $C = 90°$ directly — both methods agree.

17 [Challenging] • Target: Verifying triangle solutions with the angle sum • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$.
$||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2, \quad ||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2.$ Step 2: Solve for $\cos\theta$:
$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

18 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 8.02 Graphing Rational Functions

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

19 [Challenging] • Target: Law of Sines • Cross-Lesson: 8.03 Introduction to Limits

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

Step 1: Find angle A: $\angle A = 180^\circ - 35^\circ - 75^\circ = 70^\circ$. Step 2: Apply Law of Sines:
$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{a \sin B}{\sin A} = \frac{8\sin(35^\circ)}{\sin(70^\circ)}$. Step 3: Evaluate: $b = \frac{8(0.5736)}{0.9397} = \frac{4.5886}{0.9397} \approx \mathbf{4.88}$.

20 [Challenging] • Target: Law of Cosines • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Two sides of a right triangle are 7 cm and 10 cm. If the third side is the hypotenuse, find: (a) the exact hypotenuse length, and (b) the measure of the smaller acute angle to the nearest tenth of a degree.

(a) Exact hypotenuse: $c = \sqrt{7^2 + 10^2} = \sqrt{49 + 100} = \sqrt{\mathbf{149}}$ cm ≈ 12.21 cm. (b) Smaller acute angle: The smaller angle is opposite the shorter leg (7 cm): $\tan\theta = \frac{7}{10} = 0.7 \Rightarrow \theta = \arctan(0.7) \approx \mathbf{35.0}^\circ$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Law of Sines

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. **41.4°** ✓

b. 46.4°

c. 138.6°

d. 36.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{100 + 144 - 64}{240} \approx 0.75$, so $A = \cos^{-1}(0.75) \approx 41.4°$.

22 [Foundational] • Target: Law of Cosines

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. **44.4°** ✓

d. 39.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{36 + 49 - 25}{84} \approx 0.714$, so $A = \cos^{-1}(0.714) \approx 44.4°$.

23 [Foundational] • Target: The ambiguous case (SSA)

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120°$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. **14.8** ✓

c. 15.8

d. 12.2

By the Law of Cosines: $a^2 = b^2 + c^2 - 2bc\cos A = 10^2 + 7^2 - 2(10)(7)\cos 120° \approx 219.0$, so $a \approx 14.8$.

24 [Foundational] • Target: Choosing the right law

In ΔABC, $a = 5$, $b = 7$, and $\angle C = 60°$. Find side c (Law of Cosines).

a. $\sqrt{39}$ ✓

b. $1 + \sqrt{39}$

c. $\sqrt{74}$

d. 12

$c^2 = 5^2 + 7^2 - 2(5)(7)\cos 60° \Rightarrow c = \sqrt{39}$.

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25 [Foundational] • Target: Heron's formula

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

a. **7.2** ✓

b. 10.8

c. 8.4

d. 9.6

Let CD be the altitude from C to the hypotenuse AB in the right-angled triangle ABC. We are given AC = 9 and BC = 12. First, we find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2 = 9^2 + 12^2 = 81 + 144 = 225$. Thus, $AB = \sqrt{225} = 15$. The area of triangle ABC can be calculated in two ways: $\frac{1}{2} \times AC \times BC$ and $\frac{1}{2} \times AB \times CD$. Equating these two expressions for the area, we get.

26 [Foundational] • Target: Navigation and bearing problems

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

a. **6 units** ✓

b. 13 units

c. 36 units

d. 7.5 units

In a right-angled triangle, the altitude drawn to the hypotenuse creates three similar triangles: the original triangle and two smaller triangles. Specifically, triangle ADC is similar to triangle CDB. A property derived from this similarity is that the square of the length of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. This is known as the Geometric Mean Theorem. Therefore, $CD^2 = AD \times DB$. Given $AD = 4$ and $DB = 9$, we have.

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27 [Foundational] • Target: Solving oblique triangles from a diagram

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. Given that AC = 5, BC = 12, and angle BAC = θ, find the value of $\sin(\theta)$.

a. $\frac{12}{13}$ ✓

b. $\frac{5}{13}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

In the right-angled triangle ABC, the side opposite to angle θ (angle BAC) is BC, and the hypotenuse is AB. We are given AC = 5 and BC = 12. We can find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 5^2 + 12^2 = 25 + 144 = 169$. Therefore, $AB = \sqrt{169} = 13$. The sine of an angle in a right-angled triangle is defined as the ratio of the length of the opposite side to the length of the hypotenuse. So.

28 [Foundational] • Target: Verifying triangle solutions with the angle sum

Use the sum formula to find $\sin(30^\circ + 60^\circ)$.

a. $\frac{11}{10}$

b. $\frac{1}{2} + \frac{\sqrt{3}}{2}$

c. -1

d. **1** ✓

$\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Choosing between multiple valid approaches

In the diagram, ΔABC is a right-angled triangle with the right angle at C. The length of the hypotenuse AB is 26 units. Point D lies on AB such that CD is the altitude to the hypotenuse. If $AD = 4$ units and $BD = x$ units, find the value of x.

a. 12

b. 13

c. 16 ✓

d. 22

In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. Specifically, $\Delta ADC \sim \Delta CDB \sim \Delta ACB$. Using the property that the square of the altitude is equal to the product of the segments of the hypotenuse (geometric mean theorem), we have $CD^2 = AD \times BD$. Alternatively, we can use the similarity between ΔADC and ΔCDB: $\frac{AD}{CD} = \frac{CD}{BD}$ which leads to $CD^2 = AD \times BD$. We also know that in a right-angled triangle, the squ.

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30 [Intermediate] • Target: Law of Sines

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

a. **6 cm** ✓

b. 5 cm

c. 7 cm

d. 8 cm

In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem (or by using similarity ratios), the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into. Therefore, $AD^2 = BD \times DC$. Given $BD = 4$ cm and $DC = 9$ cm. $AD^2 = 4 \times 9$ $AD^2 = 36$ $AD = \sqrt{36}$ $AD = 6$ cm.

31 [Intermediate] • Target: Law of Cosines

The diagram shows a right-angled triangle ABC, with the right angle at C. The altitude from C to the hypotenuse AB is CD. Given that $AD = 4$ and $DB = 9$, find the length of the altitude CD.

a. **3** ✓

b. 4

c. 5

d. 6

In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles: $\Delta ABC \sim \Delta ACD \sim \Delta CBD$. Applying the geometric mean theorem for the altitude, we have $CD^2 = AD \times DB$. Given $AD = 4$ and $DB = 9$, we substitute these values into the formula: $CD^2 = 4 \times 9 = 36$. Taking the square root of both sides, $CD = \sqrt{36} = 6$. Therefore, the length of the altitude CD is 6.

32 [Intermediate] • Target: The ambiguous case (SSA)

In a triangle, sides $a = 15$ and $b = 14$ meet at an included angle of 123°. Using the Law of Cosines, which expression gives side c?

a. $\sqrt{15^2 + 14^2 + 2(15)(14)\cos 123^\circ}$

b. $\sqrt{15^2 - 14^2 - 2(15)(14)\cos 123^\circ}$

c. $\sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$ ✓

d. $15^2 + 14^2 - 2(15)(14)\cos 123^\circ$

Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C \Rightarrow c = \sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$.

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33 [Intermediate] • Target: Choosing the right law

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. **100** ✓

b. 110

c. 10

d. 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

34 [Intermediate] • Target: Heron's formula

In the right-angled triangle ABC shown below, the length of the side adjacent to angle A is 12 units and the length of the hypotenuse is 13 units. What is the cosine of angle A?

a. $\frac{5}{13}$

b. $\frac{12}{13}$ ✓

c. $\frac{5}{12}$

d. $\frac{13}{5}$

In a right-angled triangle, the cosine of an angle is defined as the ratio of the length of the adjacent side to the length of the hypotenuse. For angle A in triangle ABC, the adjacent side is given as 12 units and the hypotenuse is given as 13 units. Therefore, $\cos(A) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{13}$.

35 [Intermediate] • Target: Navigation and bearing problems

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

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36 [Intermediate] • Target: Solving oblique triangles from a diagram

A right triangle has legs 5 and 12. Hypotenuse is:

a. 11

b. **13** ✓

c. 17

d. 15

Applying Lines, Triangles & Polygons principles, $\sqrt{25+144} = \sqrt{169} = 13$ Therefore, the verified correct option is 13.

37 [Intermediate] • Target: Verifying triangle solutions with the angle sum

Find the sum of the first 5 terms of the arithmetic series with $a_1 = 1$ and $d = 4$.

a. **45** ✓

b. 49

c. 40

d. 50

$a_5 = 1 + (5-1)(4) = 17$. $S_5 = \frac{5}{2}(a_1 + a_n) = \frac{5}{2}(1+17) = 45$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 8.02 Graphing Rational Functions

Convert the polar point $(4, 30^\circ)$ to Cartesian coordinates.

a. $(2\sqrt{3}, 3)$

b. $(-2\sqrt{3}, -2)$

c. $(1+2\sqrt{3}, 2)$

d. $(2\sqrt{3}, 2)$ ✓

$x = r\cos\theta = 4\cos 30^\circ = 2\sqrt{3}$, $y = r\sin\theta = 4\sin 30^\circ = 2$.

39 [Challenging] • Target: Law of Sines • Cross-Lesson: 8.03 Introduction to Limits

Given $f(x) = (x-2)^2(x+1)$, find the multiplicity of the root $x = 2$.

a. 3

b. 1

c. **2** ✓

d. 4

The factor $(x-2)$ appears with exponent 2, so multiplicity $= 2$.

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40 [Challenging] • Target: Law of Cosines • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{1}{2}$.

a. 17

b. 20

c. 18 ✓

d. 19

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{9}{1-(\frac{1}{2})} = 18$.

41 [Challenging] • Target: The ambiguous case (SSA) • Cross-Lesson: 9.02 Geometric Sequences and Series

Find the sum of the first 3 terms of the geometric series with $a_1 = -1$ and $r = 3$.

a. -13 ✓

b. -3

c. -12

d. -14

$S_n = \frac{a_1(1-r^n)}{1-r} = \frac{-1(1-(3)^3)}{1-(3)} = -13$.

42 [Challenging] • Target: Choosing the right law • Cross-Lesson: 9.03 The Binomial Theorem

A right triangle has legs 5 and 12 with hypotenuse 13. Find $\tan\theta$, where θ is the angle opposite the side of length 5.

a. $\frac{12}{13}$

b. $\frac{13}{5}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$ ✓

$\tan\theta = \text{opposite/adjacent} = \frac{5}{12}$.

43 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Functions and Function Notation

Use the difference formula to find $\cos(90^\circ - 30^\circ)$.

a. $-\frac{1}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{3}{5}$

d. $\frac{1}{2}$ ✓

$\cos(90^\circ - 30^\circ) = \cos 60^\circ = \frac{1}{2}$.

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44 [Challenging] • Target: Navigation and bearing problems • Cross-Lesson: 1.02 Domain and Range

Find the sum of the first 10 terms of the arithmetic sequence with $a_1 = 3$ and $d = 2$.

a. 130

b. 118

c. 21

d. **120** ✓

$S_{10} = \frac{10}{2}(2(3) + (10 - 1)(2)) = 120.$

45 [Challenging] • Target: Solving oblique triangles from a diagram • Cross-Lesson: 1.03 Average Rate of Change

Find the sum of the first 5 terms of the geometric sequence with $a_1 = 2$ and $r = 3$.

a. **242** ✓

b. 162

c. 240

d. 245

$S_5 = \frac{2(1 - (3)^5)}{1 - (3)} = 242.$

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